LESSON PLAN

MOCKUP SQUADRON ENVIRONMENT (MSE) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	21	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Demonstration and explanation on the operation of various elements of mechanics of an Altimeter, ASI, and VSI & Mach meter.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Demonstration and explanation on the operation of various elements of mechanics of an Altimeter, ASI, and VSI & Mach meter.
5. Specific Objectives	24.4.1 Trainee will be able to explain the operation of Altimeter, ASI, and VSI & Mach meter.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of instrument mechanics using cutaway models; Safety: Handle glass components with care; Supervision: Instructor monitors trainee interaction with models.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an altimeter?
(b) How does an Air Speed Indicator measure speed?
(c) What is the relationship between vertical speed and pressure changes?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Static pressure system operation • Altimeter case and capsule mechanics • Kollsman window adjustment function • Altitude setting and pressure reference • Error sources in altimeter readings	L & Ds	PPT, IB	15 min
(b)	• ASI pitot tube and vent operation • Diaphragm and Bourdon tube mechanics • Calibration and error correction • Impact of airspeed on instrument accuracy • Relationship between indicated and true airspeed	L & Ds	PPT, PDF	20 min
(c)	• VSI diaphragm and capillary tube action • Rate of climb and descent indication • Mach meter operation and compressibility • Calibration of vertical speed indicators • Integration of Mach number with airspeed	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the static port affects altimeter accuracy.
(b) Describe the mechanical difference between an ASI and a VSI.
(c) How does the Mach meter account for compressibility effects?

12. Home Assignment and Follow Up	Prepare a brief note comparing the static pressure usage in the Altimeter versus the ASI for the next lesson (LP 02).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Demonstration of construction and explanation on operation of Directional indicator, Artificial Horizon & TSI.	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min